

27/08/26

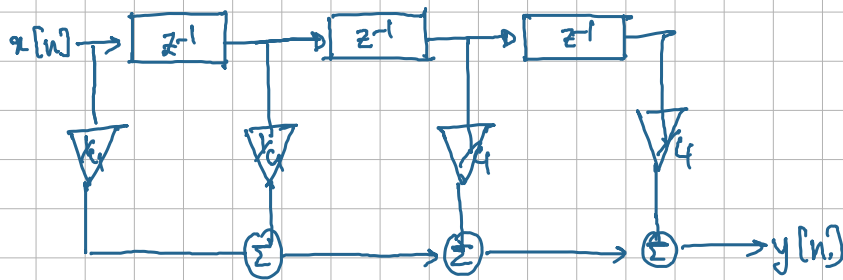
Signal Processing

- Recap: LTI Systems
- Demo: Low pass filter
- DTFT
- Examples
- Recap: An LTI system $T\{\}$ satisfies linearity and time-invariance. This leads to the convolution sum for an arbitrary input $x[n]$
$$y[n] = \sum_{k=-\infty}^{\infty} x[k] \cdot h[n-k], \text{ where } h[n] = T\{\delta[n]\}.$$
- $Y(e^{j\omega}) = X(e^{j\omega}) \cdot H(e^{j\omega})$
- Revisit the averaging system $h_{\text{ave}}[n] = \frac{1}{2} [\delta[n] + \delta[n-1]]$, a length 2 filter
- $h_{\text{ave}}[n] = \frac{1}{4} [\delta[n] + \delta[n-1] + \delta[n-2] + \delta[n-3]]$
- Boolean Board:
 - Spartan Board
 - Logic slices
 - DSP slices
 - Block RAM
 - Clock management filter
 - Analog to digital conversion (12 bit)
 - Config mem
 - Package:
 - Input: 12 bit ADC, potentiometer, 16 slide switches, 4 push buttons
 - Output: 16 discrete LEDs, 2 RGB LEDs, 8-digit seven segment display, audio, video
 - Actuators: Motor
 - DSP Slice:
 - Multiplier + Accumulator directly without LUT. Hardwired.

Vivado Design Flow:

- HDL Entry $\rightarrow$ Behavioral Simulation $\rightarrow$ Synthesis $\rightarrow$ Implementation $\rightarrow$ Bitstream $\rightarrow$ Hardware Validation.

Filter architecture:



$$h_{\text{avg}}[n] = \frac{1}{4} (\delta[n] + \delta[n-1] + \delta[n-2] + \delta[n-3])$$

FPGA implementation on Vivado.

MATLAB implementation.

Vivado

Clock tick is 10 ns

Let $x[n]$ be the clean signal

Let $\tilde{x}[n]$ be the observed noisy signal.

$$\tilde{x}[n] = x[n] + \eta[n]$$

$$\tilde{X}(e^{j\omega}) = \underbrace{X(e^{j\omega})}_{\text{pure tone}} + \underbrace{\eta(e^{j\omega})}_{\text{noise spectrum}}$$

$$Y(e^{j\omega}) = \tilde{X}(e^{j\omega}) \cdot H(e^{j\omega})$$

